
Course Section: Electricity – Internal Resistance

2022 Ph H2 Q12

(a)

Calculate the current in the circuit

Use Ohm's Law:

$$V = IR \quad \Rightarrow \quad I = \frac{V}{R}$$

Given:

- $V = 1.2 \text{ V}$ (terminal potential difference)
- $R = 15 \Omega$ (external resistor)

$$I = \frac{1.2}{15} = 0.080 \text{ A}$$

(b)

Calculate the internal resistance of the power supply

Use the relationship:

$$E = V + Ir$$

Where:

- $E = 1.5 \text{ V}$ (emf)
- $V = 1.2 \text{ V}$
- $I = 0.080 \text{ A}$

Rearrange to find internal resistance:

$$r = \frac{E - V}{I} = \frac{1.5 - 1.2}{0.080} = \frac{0.3}{0.080} = 3.75 \Omega$$

Revision Tips

- Terminal voltage is always **less than emf** when current flows, due to internal resistance
- Use $E = V + Ir$ or $V = E - Ir$ to solve internal resistance problems
- Don't confuse emf (maximum possible voltage) with terminal voltage (actual voltage across external resistor)
- Be precise: internal resistance **wastes energy** as heat inside the source